

AI Smart Room Automation

Mini-project-I Report

Submitted in partial fulfillment of the requirements
For Bachelor of Technology Degree of Engineering in

Computer Science & Engineering (AIML)

by

Bhide Armaan Amit (23AM1122)
Bhosale Harshwardhan Santosh (23AM1013)
Devkar Siddharth Nitin (23AM1101)
Patel Hamza M. Zaid (23AM1073)

Supervisor

Dr. Gargi Phadke



D Y PATIL
DEEMED TO BE
— UNIVERSITY —
RAMRAO ADIK
INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Department of Computer & Science Engineering (AIML)
Ramrao Adik Institute of Technology,
Sector 7, Nerul , Navi Mumbai
(Under the ambit of D. Y. Patil Deemed to be University)
April 30, 2025



D Y PATIL
DEEMED TO BE
— UNIVERSITY —
RAMRAO ADIK
INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Ramrao Adik Institute of Technology
(Under the ambit of D. Y. Patil Deemed to be University)
Dr. D. Y. Patil Vidyanagar, Sector 7, Nerul, Navi Mumbai 400 706.

Certificate

This is to certify that, the Mini-project-I titled

“AI Smart Room Automation ”

is a bonafide work done by

Bhide Armaan Amit (23AM1122)
Bhosale Harshwardhan Santosh (23AM1013)
Devkar Siddharth Nitin (23AM1101)
Patel Hamza M. Zaid (23AM1073)

and is submitted in the partial fulfillment of the requirement for the
degree of

Bachelor of Technology
Computer Science & Engineering (AIML)
to the
D. Y. Patil Deemed to be University.

Dr. Gargi Phadke
Supervisor

Dr. Gargi Phadke
Project Coordinator

Dr. Sangita Chaudhari
Head of Department

Dr. Mukesh Patil
Principal

Approval for B.Tech Mini-Project-I

This is certify the dissertation entitled “**AI Smart Room Automation**” is a bonafide work done by **Bhide Armaan Amit (23AM1122)**, **Bhosale Harshwardhan Santosh (23AM1013)**, **Devkar Siddharth Nitin (23AM1101)** and **Patel Hamza Mohammed Zaid (23AM1073)** under the supervision of **Dr. Gargi Phadke**. This dissertation has been approved for the award of Mini Project in Computer Application and science Engineering, D. Y. Patil Deemed to be University.

Examiners:

1: _____

2: _____

Supervisor:

1: _____

Date : _____

Place : _____

Declaration

I declare that this written submission represents my ideas in my own word and where others ideas or words have been included, I have adequately cited and referenced the original source. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of above will be cause for disciplinary action by institute and can also evoke penal action from the source which have thus not been properly cited or from whom proper permission has not been taken when needed.

Bhide Armaan Amit (23AM1122)

Bhosale Harshwardhan Santosh (23AM1013)

Devkar Siddharth Nitin (23AM1101)

Patel Hamza M. Zaid (23AM1073)

Date : _____

Acknowledgement

We, the project team, would like to express our sincere gratitude to everyone who contributed to the successful completion of our mini project. First and foremost, we are thankful to our **Principal, Dr. Mukesh Patil**, for providing us with the opportunity and support to carry out this project. We extend our heartfelt thanks to our **Head of Department, Dr. Sangita Chaudhari**, for the encouragement and continuous support throughout the project. We are especially grateful to our **Supervisor, Dr. Gargi Phadke**, for their valuable insights, consistent guidance, and support that greatly helped us throughout the project.

We would like to acknowledge the motivation and support from our friends and classmates, who have been helpful throughout the course of the project. A heartfelt thanks to our parents and siblings, whose constant love, patience, and encouragement inspired us to give our best. Lastly, we are grateful to everyone who directly or indirectly supported us in the successful completion of this project.

Abstract

In today's world, the use of smart technology in daily life is becoming increasingly common. With the growing need for energy efficiency, convenience, and security, smart room automation has become an important area of focus. This project, titled *AI Smart Room Automation*, aims to develop a system that automatically controls appliances like lights, fans, and air conditioners based on whether a person is present in the room. The goal is to reduce electricity usage and improve user comfort using Artificial Intelligence (AI) and Internet of Things (IoT) technologies.

The system works by using a camera to monitor the room. A computer vision model, such as YOLO or OpenCV, processes the video feed to check if a human is detected. If someone enters the room, the system automatically turns on the required appliances. If no one is detected for a certain amount of time, the appliances are turned off. This prevents unnecessary power consumption.

All processing is done through a cloud server. Once the AI model confirms the presence or absence of a person, the information is sent to the cloud. The cloud server then decides what actions need to be taken and sends control signals to IoT-connected appliances. These signals are received by microcontrollers like ESP32, which then switch the devices on or off.

The system can also be monitored remotely. A web interface allows users to check the status of their devices and control them even when they are not at home. This adds an extra layer of flexibility and user control.

A unique feature of this system is its ability to divide the video feed into two zones: left and right. This means that the room is treated as two separate areas. The AI model checks both halves of the video frame to determine if a person is present on the left side, the right side, or both. This makes the system more intelligent and efficient, as it can control appliances only in the occupied part of the room rather than the entire space.

For example, if a person is sitting or moving on the right side of the room, only the lights and fan on that side will be turned on. If both sides are occupied, appliances on both sides will be activated. When no presence is detected on one or both sides, the system turns off the devices in those areas accordingly. This targeted control helps save more energy and improves the user experience by responding more accurately to real-time conditions. The system is designed to be modular and flexible. It can easily be expanded to control more devices or cover larger spaces, such as multi-room homes or office environments. Additional sensors or smart devices can be added without changing the core architecture. This makes the system future-ready and adaptable to different user needs and spaces.

In real-life usage, this project has the potential to greatly improve the way we interact with our surroundings. Whether it's a home where the system adjusts lighting based on where the person is sitting, or an office where unused meeting rooms are automatically powered down, the benefits in terms of energy savings and user convenience are signifi-

cant. Over time, as AI models continue to learn and adapt to user habits, the system can become even more responsive and personalized.

One of the main benefits of this system is its ability to save energy by ensuring that electrical devices are only active when needed. It also offers improved convenience by eliminating the need for manual switches. The system is scalable, meaning it can be used in different environments such as homes, offices, schools, hospitals, and hotels.

This project shows how AI and IoT can be combined to create a smart, efficient, and user-friendly room automation system. It is easy to use, affordable to set up, and makes everyday life more comfortable while supporting energy-saving goals. With its focus on automation, modularity, and intelligent behavior, the AI Smart Room Automation system represents a meaningful step toward the future of smart living.

Keywords:

IoT, Systems, Relay, ESP32, YOLO, OpenCV

Contents

Abstract	v
List of Figures	viii
List of Tables	ix
1 Introduction	1
1.1 Motivation	2
1.2 Objective	3
1.3 Scope for work	4
1.4 Organization of Topics	4
2 Literature Survey	6
3 Proposed Method: AI Smart Room Automation	8
3.1 Overview	8
3.2 Proposed Method	8
3.3 Components	10
3.3.1 Microcontroller: ESP32	11
3.3.2 Temperature Sensor: DHT11	12
3.3.3 Camera	13
3.3.4 Relay	14
3.3.5 Server	15
3.4 Software	16
3.4.1 YOLO: You Only Look Once	16
3.4.2 OpenCV	17
3.4.3 Packages	19
3.5 Working Principle	21
4 Result	25
5 Conclusion	31
5.1 Application: Smart Paking Mangement	32
References	33

List of Figures

3.1	Block Diagram: Proposed Method: AI Room Automation	9
3.2	Microcontroller: ESP32	11
3.3	Temperature Sensor: DHT11	13
3.4	Capture Device: Camera	14
3.5	Relay Module	15
3.6	Flow Chart: Working Principle	22
4.1	Proposed Method: Working Setup	27
4.2	Classroom Monitor: Human Present Left Side	28
4.3	Left Devices ON	28
4.4	Classroom Monitor: Human Present Right Side	29
4.5	Right Devices ON	29
4.6	Classroom Monitor: Human Present And Temperature>28	30
4.7	All Devices ON	30

List of Tables

1.1	Comparison Between Traditional and AI-Based Room Automation	2
3.1	Key Components Used in the AI Smart Room Automation System	10

Abbreviations

IoT	Internet of Things
ML	Machine Learning
AI	Artificial Intelligence
YOLO	You Only Look Once
ESP32	Espressif System Protocol 32-bit
DHT11	Digital Temperature And Humidity Sensor
OpenCV	Open Source Computer Vision Library

Chapter 1

Introduction

In recent years, the emergence of Artificial Intelligence (AI) and the Internet of Things (IoT) has transformed how we interact with our living spaces. The demand for smart, connected, and energy-efficient environments has grown rapidly, especially in homes, offices, and institutions. Traditional automation systems often depend on manual operation or basic motion sensors, which can be inefficient and lack the ability to adapt to real-time human behavior. As a result, these systems may lead to energy waste and do not always provide the desired level of comfort or flexibility.

The AI Smart Room Automation project aims to address these issues by combining AI-powered human detection with cloud-based IoT control. The goal is to create an intelligent, responsive, and energy-saving system that can automatically manage room appliances like lights, fans, and air conditioners. Instead of relying on fixed schedules or manual switches, the system reacts in real time to human presence and activity.

A camera is used to monitor the room continuously. This video feed is analyzed using AI models such as OpenCV, YOLO, and TensorFlow, which can accurately detect whether someone is present. Once human presence is confirmed, the information is sent to a cloud server where decision-making logic is applied. Depending on the situation, the system can turn appliances on or off automatically. This reduces electricity consumption and ensures that appliances are only used when needed.

An advanced feature of this system is that it divides the video feed into two zones: left and right. The AI model checks each side separately to see if a person is present. If someone is only on the right side, only the appliances on that side are turned on. If both sides are occupied, the system activates appliances on both sides. This targeted control not only

Table 1.1: Comparison Between Traditional and AI-Based Room Automation

Parameter	Traditional Automation	AI Smart Room Automation
Control Mechanism	Manual switches or motion sensors	AI-based human detection with cloud-based IoT control
Energy Efficiency	Moderate; appliances may stay on unnecessarily	High; appliances turn off automatically when room is vacant
Accuracy	Limited to basic motion detection	High accuracy using computer vision (YOLO + OpenCV)
Customization	Relies on timers or fixed presets	Dynamically adapts to user presence and zone detection
Environmental Adaptability	Lacks integration with real-time conditions	Responds to temperature input and occupancy in real time

improves efficiency but also offers a more intelligent and personalized user experience. The cloud-based design allows for easy updates and integration with other smart systems. Appliances are controlled through IoT-enabled microcontrollers such as ESP32 and the entire system can be managed remotely via a mobile app or web interface. This ensures that users can check or control their devices anytime, from anywhere.

1.1 Motivation

The motivation behind the AI Smart Room Automation system comes from the increasing demand for intelligent, efficient, and user-friendly living environments. As technology continues to evolve, people expect their homes and workplaces to become more responsive, automated, and energy-conscious. Traditional home automation systems often rely on manual switches or basic motion sensors, which can lead to inefficient use of electricity and limited functionality. These systems do not adapt well to changing user behavior and often require unnecessary user interaction. This project aims to solve these challenges by combining artificial intelligence with cloud-based IoT control. By using real-time video data and AI models to detect human presence, the system can automatically control ap-

pliances such as lights, fans, or air conditioners. This creates a smarter and more seamless experience for users, where appliances are used only when needed — without the need for manual input. The system also helps reduce energy waste, which is especially important in today's world where sustainable living is a global priority. By detecting when a room is unoccupied and turning off devices accordingly, the system can significantly lower electricity consumption. At the same time, it improves comfort by instantly adapting the room environment to match user activity.

Key Motivations:

Energy Conservation: Automatically switch off appliances when no one is present to minimize unnecessary power usage.

Hands-Free Convenience: Remove the need for manual switches through real-time AI-based automation.

Smart Living: Build an environment that responds intelligently to where users are and what they're doing.

1.2 Objective

The AI Smart Room Automation project aims to develop an intelligent and energy-efficient system that uses artificial intelligence for human detection and cloud-based IoT control to manage room appliances. The goal is to make daily life more convenient while reducing electricity usage through smart, responsive automation.

The specific objectives of the project are as follows:

To automate room appliances such as lights, fans, and air conditioners based on real-time human presence. To apply AI-based human detection techniques using models like YOLO and OpenCV for accurate occupancy recognition. To integrate cloud computing with IoT-enabled devices for real-time remote monitoring and control. To reduce electricity consumption by automatically turning off appliances when not needed. To create a smart, user-friendly system that adapts to activity within the room without requiring manual operation. By achieving these objectives, the system supports the development of a smarter, more sustainable, and comfortable living environment while minimizing energy waste and manual effort.

1.3 Scope for work

This project focuses on the design, development, and testing of an AI-based smart room automation system that combines computer vision, cloud computing, and IoT technologies. The system is built to improve user comfort, reduce energy consumption, and enable intelligent appliance control through real-time human presence detection.

The scope of the system includes the following key components and functionalities:

AI-Based Human Detection: Implement real-time human detection using computer vision techniques such as YOLO, OpenCV, and TensorFlow. A live video stream is used to monitor occupancy and trigger automation events.

Cloud-Based Processing: Use cloud infrastructure to process detection data, run logic-based decision-making algorithms, and send control signals to connected IoT devices. This enables centralized processing and remote accessibility.

IoT Device Integration: Manage electrical appliances like lights, fans, and air conditioners through microcontrollers (e.g., ESP32). These devices receive cloud instructions over communication protocols such as HTTP or MQTT.

Remote Monitoring and Control: Provide users with a web dashboard² that allows them to view the system status and control appliances remotely, enhancing user flexibility and convenience.

Energy Efficiency Optimization: Conserve electricity by ensuring that devices are only powered when needed. Appliances are turned off automatically when no occupancy is detected, supporting a more sustainable and cost-effective environment.

Scalability and Flexibility: Develop a modular and extensible system architecture that can be easily expanded to multiple rooms, buildings, or integrated with other smart home platforms and sensors in the future.

The scope is designed to ensure the system is functional, adaptable, and future-ready, making it suitable for a range of environments such as homes, offices, institutions, and other smart spaces.

1.4 Organization of Topics

This report is structured into six chapters, each focusing on a distinct aspect of the AI Smart Room Automation system. The organization ensures a logical progression from

the conceptual foundation to the technical implementation and evaluation.

Chapter 1: Introduction

Provides the context and background of the project, highlighting the growing need for intelligent automation in modern environments. It includes a literature review of current trends, the motivation behind the project, its objectives, scope, and a brief outline of the topics covered in the report.

Chapter 2: Literature Survey

Presents a comprehensive review of how the project was made, it takes us through the development process step by step.

Chapter 3: Proposed Method – AI Smart Room Automation

Describes the core methodology of the system. It is subdivided into sections detailing the architectural overview, the step-by-step workflow, key system components, and the working principle of the automation process. This chapter forms the heart of the report.

Chapter 4: Result

Summarizes the outcome of the implemented system, including its performance in real-world testing scenarios.

Chapter 5: Conclusion

Reflects on the overall contribution of the project, highlighting the value added by AI-based automation in smart environments. It also covers practical applications and presents a vision for future development and integration.

References

Lists all the academic and technical sources cited throughout the document, adhering to IEEE citation standards.

Chapter 2

Literature Survey

Smart room automation, enabled by Artificial Intelligence (AI) and the Internet of Things (IoT), has emerged as an essential component in today's smart homes and industrial settings. The combination of AI-driven human detection with IoT-enabled device control represents a significant advancement in developing adaptable settings that maximize energy savings, security, and user ease.

AI in Human Detection and Room Monitoring

Human detection using real-time video analysis is an essential component of AI Smart Room Automation. The models used, such as YOLO (You Only Look Once) and OpenCV, have shown significant performance in detecting human presence in indoor environments: The authors Redmon, J., & Farhadi, A.[20] in their paper introduce the YOLO algorithm, which is commonly used for object recognition in real-time systems. YOLOv3 achieves a balance of speed and accuracy, making it ideal for smart room applications. The authors Jiang, Peiyuan and Ergu, Daji and Liu, Fangyao and Cai, Ying and Ma, Bo[11], in their paper have done a through analysis on YOLO and have given many insightful remarks. The authors Terven, Juan and Córdova-Esparza, Diana-Margarita and Romero-González, Julio-Alejandro[25], in their works have given a comprehensive analysis of YOLO's evolution, examining the innovations and contributions in each iteration from the original YOLO up to YOLOv8, YOLO-NAS, and YOLO with transformers.

Cloud-Integrated IoT for Automation

The combination of cloud computing with IoT provides centralized decision-making and remote control of devices: The authors Atzori, L., Iera, A., & Morabito, G.[5] in their foundational work outline the structure of IoT-based systems and emphasize the role

of cloud services in scalability and interoperability. The authors Stolojescu-Crisan, C., Crisan, C., & Butunoi, B. P. [22] in their research present a real-world implementation of IoT and cloud services for home automation, closely aligning with the project's method. The authors Malik, Akash and Om, Hari[14], in their paper give the introduction of cloud and IoT, their integration architecture, integration applications, and challenges and issues involved are discussed. The authors Maier, Alexander and Sharp, Andrew and Vagapov, Yuriy[13], in their work provide a comparative analysis of the ESP32 with some other market competitors and introduces the microcontroller specification, features and programming details.

Machine Learning for Automation Decision-Making

Instead of fixed rules, AI-based systems employ ML models for adaptive responses:

The author Akram, Ali, [2] in his work proposes a self-learning mechanism that optimizes energy usage through human activity predictions. The author Park S. [18] in his work introduces adaptive scheduling strategies that can be embedded in cloud-based home systems. The author Hüllermeier, Eyke[9], in his work elaborates on specific characteristics of prescriptive ML and highlights some key challenges it implies drawing connections to other branches of contemporary AI research, the grounding of prescriptive ML in a (generalized) decision-theoretic framework is advocated. The authors Sujatha, M and Priya, N and Beno, A and Blesslin Sheeba, T and Manikandan, M and Tresa, I Monica and Jose, P Subha Hency and Peroumal, Vijayakumar and Timothy, Sojan Palukaran[23] in their work use the decision tree algorithm is for monitoring energy consumption in machines and appliances, predicting future behaviour, and detecting anomalous behaviour. The authors Adekunle, Bolaji Iyanu and Chukwuma-Eke, Ezinne C and Balogun, Emmanuel Damilare and Ogunsola, Kolade Olusola[1] in their work explore the role of ML in automation, focusing on its applications in predictive maintenance, quality control, supply chain optimization, and real-time monitoring.

Chapter 3

Proposed Method: AI Smart Room Automation

3.1 Overview

The AI Smart Room Automation system is designed to intelligently control room appliances based on human presence using a combination of AI, cloud computing, and IoT[12]. The system uses a camera to capture real-time video of the room, where an AI model processes the footage to detect human occupancy. Based on the detection, decisions are made in the cloud and appropriate commands are sent to IoT devices like lights, fans, and air conditioners.

3.2 Proposed Method

The AI Smart Room Automation system is designed to detect human presence in a room using computer vision and AI models, analyze the detection through a cloud-based server, and control connected IoT devices (like lights, fans, or AC) accordingly. This three-tier system—comprising sensing, decision-making, and actuation—ensures energy efficiency, user convenience, and scalability.

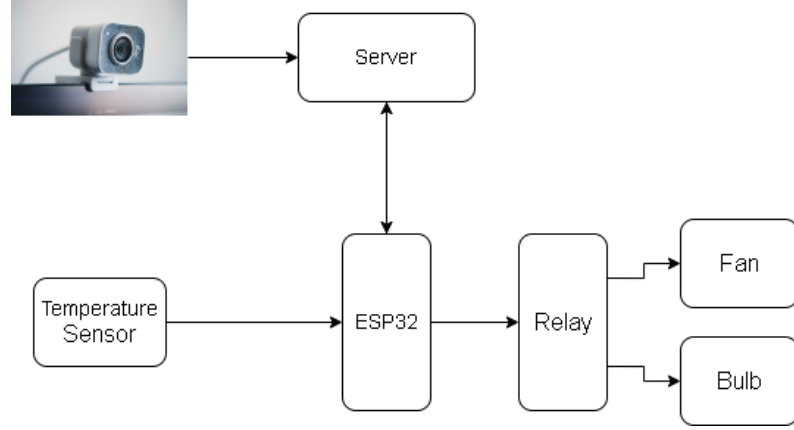


Figure 3.1: Block Diagram: Proposed Method: AI Room Automation

3.1 The block diagram above illustrates the overall working of the AI Smart Room Automation system. It integrates components such as a camera, temperature sensor, cloud server, ESP32 microcontroller, relay module, and electrical appliances like a fan and bulb. The design focuses on enabling intelligent automation through real-time human presence detection and environmental sensing.

The camera serves as the primary input device for monitoring room occupancy. It captures a live video feed of the room, which is continuously sent to the server for processing. The server acts as the brain of the system. It hosts AI models (such as YOLO or OpenCV) that analyze the camera feed in real time to detect human presence. Once the presence is detected and evaluated, the server sends corresponding control signals to the ESP32 based on predefined logic. In addition to human detection, a temperature sensor is used to monitor the room's ambient temperature. This sensor sends real-time temperature data directly to the ESP32, allowing the system to make decisions about activating appliances like fans based on thermal conditions. The ESP32 microcontroller acts as the communication bridge between the server and the physical appliances. It receives control instructions from the server (e.g., turn on/off appliances based on occupancy) and also takes input from the temperature sensor to make additional decisions locally. The relay module is used to switch high-voltage devices like fans and bulbs. The ESP32 controls the relays through its GPIO pins. Depending on the command received, the corresponding relay is activated or deactivated, thus powering ON or OFF the connected appliances. These represent the room appliances being controlled in the system. Based on the occupancy data and temperature input, the system intelligently decides whether to activate the fan

or bulb, optimizing energy usage and user comfort.

3.3 Components

The AI Smart Room Automation system is built using a combination of hardware and software components that work together to detect human presence, process environmental data, and automate appliances intelligently. Each component in the system plays a critical role in ensuring smooth communication between the user, cloud server, AI model, and physical devices.

This section provides a detailed overview of the major components used in the project. By integrating these components, the system is able to operate in real time, respond to occupancy and environmental conditions, and execute control commands accurately. The selection of each component was based on functionality, compatibility, cost-effectiveness, and reliability in smart automation environments.

Table 3.1: Key Components Used in the AI Smart Room Automation System

Component	Technology Used	Function
Camera Module	USB Webcam	Captures real-time video feed from the room to monitor occupancy
AI Detection Model	YOLOv8 with OpenCV	Processes video frames to detect human presence and determine position (left/right)
Microcontroller	ESP32 with Web-Server.h	Receives HTTP commands and controls appliances via relay module
Relay Module	4-Channel 5V Relay	Switches electrical appliances (lights/fans) ON or OFF based on ESP32 commands
Environmental Sensor	Temperature Sensor(DHT11)	Measures room temperature and sends data to ESP32 for condition-based automation

3.3.1 Microcontroller: ESP32

The ESP32 is a low-cost, low-power system-on-chip microcontroller with integrated Wi-Fi and Bluetooth capabilities[16]. In the AI Smart Room Automation project, ESP32 plays a vital role in bridging the gap between cloud-based decision-making and the physical control of room appliances. It serves as the IoT controller responsible for receiving instructions from the cloud server and executing the required operations on devices such as lights, fans, or air conditioners.

One of the key reasons for using ESP32 in this project is its excellent support for wireless communication via Wi-Fi. This allows the microcontroller to maintain a continuous connection with the cloud, where AI-based human detection decisions are made. Once a decision is computed—such as turning on a light or switching off a fan—it is sent over the network to the ESP32, which interprets the command and controls the corresponding hardware components through its GPIO (General Purpose Input/Output) pins.

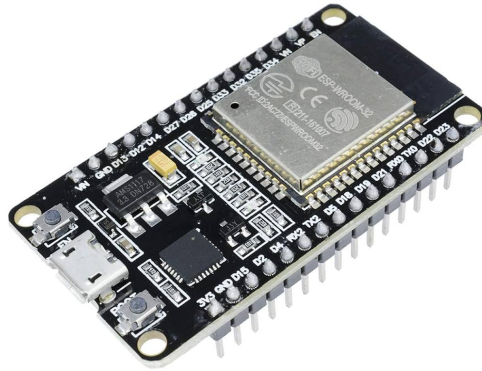


Figure 3.2: Microcontroller: ESP32

ESP32 is a single 2.4 GHz Wi-Fi-and-Bluetooth combo chip designed with the TSMC low-power 40 nm technology. It is designed to achieve the best power and RF performance, showing robustness, versatility and reliability in a wide variety of applications and power scenarios.

In our project, the ESP32 microcontroller plays a crucial role in controlling room appliances based on real-time decisions made by the cloud. It serves as the physical IoT device that receives commands from the cloud server and interacts directly with connected appliances such as lights and fans. By using the ESP32, we are able to wirelessly control hardware components in response to human detection processed through AI algorithms.

We have implemented the ESP32 in a modular way, where it handles both left and right zones of the room. This zoning is based on the AI model dividing the video frame and determining which side of the room is occupied. The cloud logic then instructs the ESP32 to turn on only the appliances on the occupied side. For instance, if the right zone is occupied, the ESP32 will turn on only the fan and light connected to the right-side relays.

3.3.2 Temperature Sensor: DHT11

The DHT11 is a widely used digital temperature and humidity sensor that is both cost-effective and easy to integrate with microcontroller-based systems. In smart automation projects like AI Smart Room Automation, the DHT11 adds an environmental sensing capability that complements the system's AI-based human detection and appliance control[6]. While the camera detects the presence of a person, the DHT11 provides real-time information about the room's temperature and humidity, enabling smarter and more adaptive automation decisions.

The DHT11 sensor contains a thermistor and a capacitive humidity sensor, which measure temperature and relative humidity respectively. It communicates using a digital signal over a single data line, which makes it very efficient and straightforward to use in microcontroller setups like ESP32. The data it provides is sufficient for indoor environmental monitoring and is accurate enough for general room control applications.

In a smart room setup, temperature data from the DHT11 can be used to determine whether to turn on a fan or an air conditioning unit. For example, if the temperature exceeds a preset threshold, the system can trigger a response to activate cooling appliances. While the DHT11 also supports humidity monitoring, many basic systems use only the temperature data for practical control purposes, especially in small-scale or prototype projects.

In our AI Smart Room Automation project, the DHT11 sensor is used to measure the room's temperature in real time and send this information to the ESP32 microcontroller. The DHT11 plays an important supporting role in making the system more responsive to environmental conditions, allowing it to not only react to human presence but also to the thermal comfort level of the room.

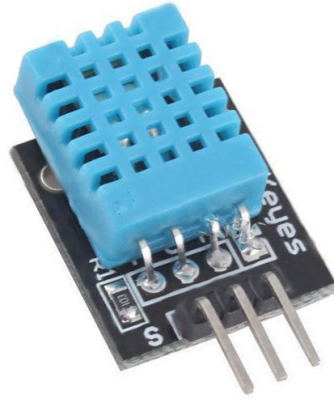


Figure 3.3: Temperature Sensor: DHT11

3.3 The DHT11 is a commonly used Temperature and humidity sensor that comes with a dedicated NTC to measure temperature and an 8-bit microcontroller to output the values of temperature and humidity as serial data.

The DHT11 is a low-cost digital sensor that provides temperature readings in degrees Celsius. It is directly connected to the ESP32 through a digital I/O pin. In our implementation, the sensor sends temperature data to the ESP32 at regular intervals. This data is processed locally by the ESP32 and is used as an input to control appliances like the fan. For instance, if the temperature crosses a predefined threshold (e.g., 28°C), the ESP32 can turn on the fan even if human presence was not recently detected. This adds an extra layer of automation based on environmental sensing.

3.3.3 Camera

In AI-based automation systems, the camera is one of the most critical components, especially when the goal is to recognize human presence and trigger intelligent control of appliances. In the AI Smart Room Automation project, the camera acts as the primary input device for visual sensing. It captures a real-time video feed of the room, which is then analyzed using computer vision algorithms to detect the presence of a person.

Unlike traditional automation systems that use motion sensors or infrared detectors, a camera-based system provides much more detailed and flexible input. It can capture the entire room view and allow the AI model to analyze not just motion but specific objects in this case, human figures. This approach allows for more accurate and context-aware decision-making, which is ideal for smart, adaptive environments.



Figure 3.4: Capture Device: Camera

3.4 The video stream captured by the camera is passed to an AI model, such as YOLO for processing.

The camera's continuous monitoring capability makes it ideal for real-time human detection applications. It does not just detect the presence of someone in a predefined area but can be used to divide the frame into zones (e.g., left and right side of the room), which enables selective control of appliances based on where the person is located. Cameras are also advantageous because they allow the use of advanced AI techniques, including pose estimation, multiple-person tracking, gesture recognition, and more — making them a future-proof option for expanding system capabilities.

3.3.4 Relay

The relay module is an essential component in any smart automation system that involves controlling high-voltage electrical appliances using low-voltage signals. In the AI Smart Room Automation project, the relay module serves as the interface between the microcontroller (ESP32) and high-power devices such as lights, fans, or air conditioners. It allows the low-power digital output from the ESP32 to switch ON or OFF standard household appliances safely and efficiently.

A relay is an electrically operated switch that uses an electromagnet to mechanically operate a switch. The relay module used in most IoT projects typically consists of one or more relays along with supporting components like transistors, diodes, and optocouplers for safe isolation between the control circuit (low voltage) and the load circuit (high voltage). This makes it suitable for switching AC appliances without exposing the micro-

controller to direct high-voltage current.



Figure 3.5: Relay Module

A relay is an electrically operated switch. It consists of a set of input terminals for a single or multiple control signals, and a set of operating contact terminals. The switch may have any number of contacts in multiple contact forms, such as make contacts, break contacts, or combinations thereof.

The relay module used in this project is a 4-channel 5V relay board, which allows multiple appliances to be controlled independently. Each relay can be triggered using one GPIO pin of the ESP32, which sends a HIGH or LOW signal to toggle the switch. For example, when the AI system detects human presence and the cloud decides to turn on a light, it sends a command to the ESP32, which in turn activates the corresponding relay channel. The relay closes the circuit and powers the light.

In smart automation systems, relays are ideal because they are electrically isolated, durable, inexpensive, and widely available. They enable safe integration of microcontroller logic with powerful real-world devices that operate at 220V AC or higher. This functionality is key to bridging the gap between software intelligence (AI and cloud decisions) and physical action (switching appliances).

3.3.5 Server

In our AI Smart Room Automation project, the `WebServer.h` library is used to create a lightweight web server hosted directly on the ESP32 microcontroller. This web server plays a key role in enabling communication between the cloud logic (or user interface) and the ESP32, which controls appliances in real-time.

The `WebServer.h` library is part of the official ESP32 Arduino core and allows the ESP32 to listen for incoming HTTP requests on a specific IP address and port. By using this library, the ESP32 can act as a local web server that handles GET or POST requests and responds with HTML content or control signals.

Using `WebServer.h` simplifies the entire process of remote communication. It removes the need for complex MQTT or database setups when only direct control is required. It also allows for quick testing and debugging using a browser or mobile app to send HTTP requests to the ESP32's IP address.

3.4 Software

The successful implementation of the AI Smart Room Automation system relies heavily on a combination of advanced software tools and frameworks. These software components enable real-time human detection, environmental data processing, and appliance control across different hardware modules.

The major software elements used in the project include computer vision libraries, deep learning models, server frameworks, and microcontroller programming environments. Each software tool was carefully selected based on performance, compatibility, and its ability to support intelligent automation. This section outlines the key software platforms and libraries employed, explaining their individual roles and how they integrate to achieve seamless communication between the camera, server, ESP32 microcontroller, and connected appliances. By leveraging these software technologies, the system is able to process live video feeds, detect occupancy with high accuracy, respond to temperature variations, and control room appliances efficiently in real-time.

3.4.1 YOLO: You Only Look Once

In the AI Smart Room Automation system, accurate and real-time human detection is a critical feature that drives the automation of room appliances. To achieve this, the project utilizes YOLO (You Only Look Once) — one of the most widely used deep learning models for real-time object detection.

YOLO is a state-of-the-art, open-source object detection algorithm designed to identify and locate multiple objects within an image or video frame. Unlike traditional methods

that apply classifiers to multiple sections of an image (such as sliding windows or region proposals)[15], YOLO reframes object detection as a single regression problem. This approach makes YOLO extremely fast and suitable for real-time applications, which is essential in automation systems that rely on live camera input.

The key idea behind YOLO is that it processes the entire image in one go. It divides the image into an $S \times S$ grid (typically 7x7 or higher), and for each grid cell, it predicts: Bounding boxes (x, y, width, height); Confidence scores; Class probabilities (e.g., person, car, dog, etc.).

In our project, we focus only on detecting humans. The YOLO model is either pre-trained on datasets like COCO (Common Objects in Context) or custom-trained on datasets containing human figures in various environments. When a person is detected in the frame, the system draws a bounding box around them and calculates the detection confidence. One of the key advantages of YOLO is its real-time performance. Traditional object detection models such as R-CNN or Fast R-CNN offer good accuracy but are often too slow for real-time processing[21]. YOLO, on the other hand, processes frames in milliseconds, making it highly suitable for live video analysis in smart automation systems.

In our project, we use a camera connected to a computer system that runs a Python script using YOLOv8 with OpenCV. The script detects human presence from a live video feed. The YOLO model processes each video frame and identifies if a human is present by drawing a bounding box around them. Based on the location of the bounding box (left or right zone), it sends a corresponding command to the ESP32 through HTTP using flask. This simple but effective setup allows our system to: Detect if a person is on the left, right, or both sides of the room Send specific appliance control commands wirelessly to ESP32 Turn on or off lights/fans depending on detected human position The YOLO detection runs entirely on the computer (not on the ESP32), and it communicates directly with the ESP32 via local IP over Wi-Fi.

3.4.2 OpenCV

OpenCV (Open Source Computer Vision Library) is a powerful and widely-used open-source computer vision and machine learning software library[24]. In our AI Smart Room Automation project, OpenCV plays a central role in capturing live video from the camera, preprocessing the video frames, and interfacing with the YOLO model to detect human

presence in real time.

OpenCV provides the tools necessary to work directly with video streams, image processing, and object detection tasks. It is written in C++ with bindings for Python, which makes it highly efficient and easy to integrate into modern AI applications. In our project, we use OpenCV in Python to capture and process the live video feed from a standard webcam.

The real-time video stream from the camera is accessed using OpenCV's `cv2.VideoCapture()` function. Each frame is read, resized, and converted into a format suitable for YOLOv8 model input. This preprocessing includes operations like resizing, normalization, and color space conversion (BGR to RGB or vice versa). Once a frame is processed, it is passed through the YOLO detection pipeline using OpenCV's Deep Neural Network (DNN) module or an external framework like Darknet. The model returns bounding boxes, class IDs, and confidence values. OpenCV is then used to draw the bounding boxes on the frame and visually indicate which zone (left or right) the person has been detected in.

The project also uses OpenCV to: Firstly, Define the boundary between left and right zones of the frame. Secondly, Track whether a person has moved from one zone to another. Thirdly, Display the processed output in a real-time window for debugging and demonstration. In addition to its image processing capabilities, OpenCV allows us to integrate the detection result with logic-based decisions. For example, if a bounding box's center lies in the left half of the frame, OpenCV helps flag it as "left occupied," which triggers an HTTP request to the ESP32 web server. Similarly, detections on the right side send a different request. OpenCV also enables frame-by-frame analysis, motion detection (if expanded), and visual annotation — making it an ideal tool for interactive and real-time smart automation systems.

OpenCV is the backbone of our project's vision system. It handles everything from camera integration to video processing and detection display. Without OpenCV, real-time human recognition and zone-based automation control would not be practically achievable. Its role in bridging hardware (camera), AI (YOLO), and system logic (HTTP control) makes it a vital component of the AI Smart Room Automation system.

3.4.3 Packages

Ultralytics:-

The ultralytics Python package is an open-source library developed by the creators of YOLOv5 and YOLOv8[10]. It provides a high-level, easy-to-use API for training, validating, and deploying state-of-the-art object detection models. This package simplifies deep learning workflows, especially for real-time computer vision tasks such as human detection, which is central to our AI Smart Room Automation project. In our implementation, we use the ultralytics package to load and run the YOLOv8 model for detecting humans in a live webcam feed. The model returns bounding boxes and class labels, allowing us to check if a "person" is present and determine whether they are on the left or right side of the room.

Threading:-

Threading allows different parts of our system[7] — such as capturing video, running YOLO-based human detection, processing temperature data, and sending HTTP commands — to run in parallel without blocking each other. This is especially important in a real-time application where delays in one part of the system can affect the responsiveness of the entire setup. In our AI Smart Room Automation project, Python's built-in threading module is used to execute multiple tasks simultaneously within the same program. In our implementation, the video capture and detection run continuously in one thread, while another thread handles sending HTTP requests to the ESP32 based on the detection result. This separation ensures that the video processing loop is not delayed by network latency or I/O operations.

Requests:-

The requests module is a popular Python library used for sending HTTP requests easily and efficiently[19]. It simplifies communication between client and server by allowing Python programs to send data over the network using standard web protocols such as GET and POST. In our AI Smart Room Automation project, the requests module plays a critical role in sending control commands from the Python script (which runs the YOLO human detection) to the ESP32 microcontroller.

Flask:-

Flask is a lightweight, open-source web framework for Python that is designed to build simple and scalable web applications and APIs[17]. It provides the tools needed to create

HTTP routes, handle client-server communication, and manage web-based interactions easily. In our AI Smart Room Automation project, Flask is used to create a local server on the computer running the YOLO detection model. This Flask server acts as an intermediary between the human detection process and the IoT control logic. After each frame is processed by the YOLO model using OpenCV and Ultralytics, Flask helps send real-time HTTP requests to the ESP32 microcontroller based on the detection results.

JSON:-

JSON (JavaScript Object Notation) is a lightweight data interchange format that is easy for humans to read and write, and easy for machines to parse and generate[8]. It is commonly used in web and IoT applications to transmit structured data between servers, clients, and microcontrollers. In our AI Smart Room Automation project, JSON is used for data formatting and communication between the YOLO-based human detection system and the ESP32 microcontroller. Instead of sending plain text or simple HTTP requests, JSON allows packaging this information in a structured way.

Cv2:-

The cv2 module is the Python interface for OpenCV (Open Source Computer Vision Library) — a powerful and widely used open-source library for real-time computer vision and image processing tasks[24]. It provides tools for capturing, processing, and analyzing images and videos, making it essential in AI-powered systems like smart automation. In our AI Smart Room Automation project, cv2 is used for capturing live video from a USB webcam, processing each video frame, and feeding those frames into the YOLOv8 model for human detection. The function cv2.VideoCapture() is used to initialize the camera stream. After each frame is captured, it is resized, normalized, or converted in color space as needed before passing to the YOLO detection model (via Ultralytics). Once a person is detected, cv2.rectangle() and cv2.putText() are used to draw bounding boxes and labels on the image for visual feedback.

Cvzone:-

Cvzone is a high-level Python module built on top of OpenCV and MediaPipe that simplifies many common computer vision tasks[24], especially for real-time AI applications. It provides user-friendly functions for tasks like face detection, hand tracking, object detection overlays, and visual enhancements — making it ideal for prototyping and building

interactive AI systems. In our AI Smart Room Automation project, cvzone is used to simplify the processing and display of detection results. After YOLO detects a person in the frame, cvzone helps draw bounding boxes, labels, and zones more efficiently. Additionally, cvzone helps with tasks like mapping detection zones (left/right side), which is essential for our logic that controls appliances based on a person's position in the room.

WiFi.h:-

The WiFi.h library is a built-in library for the ESP32 microcontroller that allows it to connect to Wi-Fi networks and establish wireless communication[4]. It is an essential part of any IoT system where devices need to interact over a local network or the internet. In our AI Smart Room Automation project, WiFi.h is used to connect the ESP32 to the local Wi-Fi network. After successfully connecting, the ESP32 can host a web server (using WebServer.h) and listen for HTTP requests sent by the Python-based detection system running Flask and YOLO. Commands such as turning ON/OFF lights or fans are received over Wi-Fi, and the ESP32 responds instantly by controlling the connected appliances through relay modules.

HTTPClient.h:-

The httpclient.h package provides a set of functions and classes designed to facilitate HTTP communication in the AI Smart Room Automation project. This package is responsible for handling HTTP requests and responses[3], enabling the smart room system to interact with external servers, APIs, or other networked devices through the HTTP protocol.

3.5 Working Principle

3.6 The flowchart represents the overall working principle of the AI Smart Room Automation system, specifically as implemented in our project. It explains how the different components — camera, YOLO model, DHT11 sensor, ESP32 microcontroller, and relay module — work together to automate room appliances based on human presence and environmental conditions.

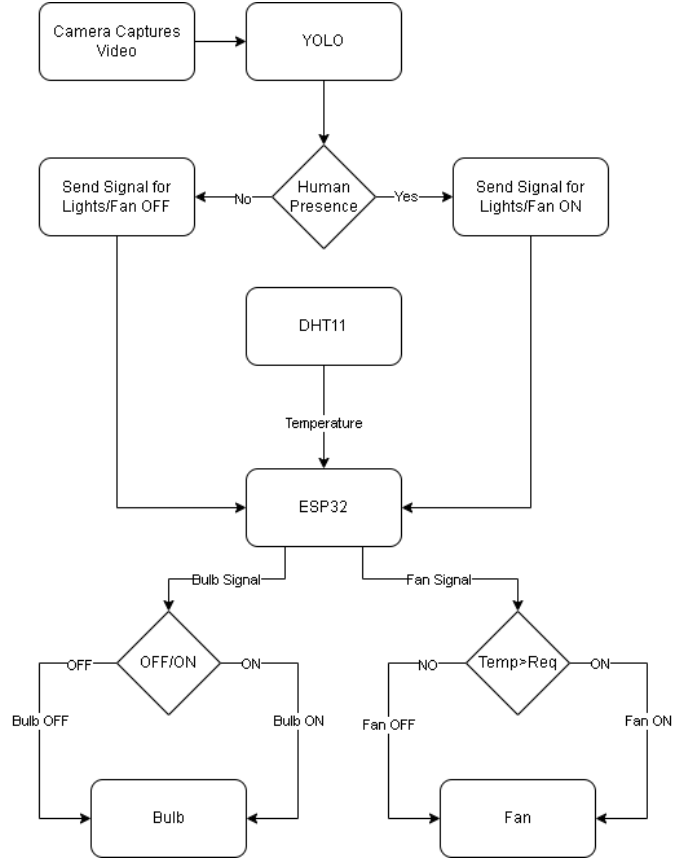


Figure 3.6: Flow Chart: Working Principle

Camera Captures Video:-

In our project, a USB webcam is installed inside the room to provide a continuous live video feed. This camera serves as the main input sensor for detecting human presence. The video stream is captured using Python's OpenCV library, which provides real-time frame-by-frame access to the feed. By constantly capturing and analyzing the environment, the camera ensures that no manual triggering is required for monitoring occupancy. Unlike traditional motion sensors, the use of a video camera allows detailed analysis, including detecting where exactly within the room a person is located (left or right zone). This high-resolution real-time input is critical for running accurate object detection models and enables a fully automated system without physical switches or user input.

YOLO-Based Human Detection:-

The live video frames captured by the camera are processed using the YOLOv8 object detection model. YOLO ("You Only Look Once") is chosen for its high-speed and reliable object detection capabilities. Using OpenCV's DNN module, the YOLO model analyzes each frame and identifies whether a human is present by generating bounding

boxes around detected objects. A confidence score is associated with each detection to minimize false positives. YOLO processes the entire frame in one go, which makes it ideal for real-time operations without lag. Based on the position of the bounding box (left or right half of the screen), the system determines which appliances to control. This fast and efficient detection pipeline ensures that the automation system responds to human presence within milliseconds.

Decision Based on Human Presence:-

After processing the frame through YOLO, the system decides whether a human is present or not. If no human is detected, the system automatically prepares to deactivate appliances like the light and fan to conserve energy. If a human is detected, it triggers a signal to activate the appliances. Communication with the ESP32 microcontroller is established using HTTP requests. For example, when a person is detected on the right side of the room, a request is sent to the ESP32. If no person is found, a request is sent to turn off lights. This simple but powerful decision logic ensures that the system only uses power when necessary, maximizing energy efficiency without requiring user involvement.

Environmental Sensing via DHT11:-

Simultaneously with the human detection process, a DHT11 sensor installed in the room measures the ambient temperature. The DHT11 sensor is directly connected to the ESP32 and operates independently of the video processing unit. It reads the temperature periodically and sends this information to the ESP32, enabling the system to make intelligent decisions about turning on or off specific appliances like the fan. If the room temperature exceeds a predefined threshold (e.g., 28°C), the fan may be turned on to improve user comfort. This dual approach — combining human detection and environmental monitoring — ensures the automation is not only responsive to occupancy but also adaptive to changing room conditions.

ESP32 Microcontroller Processing:-

The ESP32 acts as the brain of the local control system. It hosts a lightweight HTTP server using the `WebServer.h` library, allowing it to receive real-time commands from the computer that processes the YOLO detections. Upon receiving a command, the ESP32 processes the instruction and triggers the connected relays accordingly. Apart from handling external HTTP commands, the ESP32 also reads temperature data from the DHT11 sensor and makes independent decisions based on environmental conditions.

This dual-input processing capability makes ESP32 a robust and reliable controller for smart automation tasks, ensuring that the system remains functional even if one input (e.g., cloud connection) temporarily fails.

Bulb Control Logic:-

The control logic for the room's bulb is primarily driven by human detection status. If a human presence is detected and a corresponding "lights ON" command is received, the ESP32 activates the GPIO pin connected to the bulb's relay module. This closes the relay circuit and switches the bulb ON. If the room becomes empty (no human detected), a "lights OFF" command is sent to the ESP32, which opens the relay circuit and turns the bulb OFF. This real-time, occupancy-based switching mechanism ensures that lights are only active when the room is in use, drastically reducing energy waste and providing hands-free convenience for the user.

Fan Control Logic:-

Fan operation in the system is slightly more sophisticated, as it considers both human presence and the room's temperature. When the ESP32 receives a "fan ON" command, it first checks the current temperature measured by the DHT11 sensor. If the temperature exceeds a defined threshold (e.g., 28°C), the fan relay is activated, and the fan starts operating. If the temperature is below the threshold or no human is present, the ESP32 ensures that the fan remains OFF. This logic ensures not just occupancy-based control but also environmental adaptability, optimizing comfort while maintaining energy efficiency.

Relay and Appliance Actuation:-

Relays serve as electronic switches that enable the low-power ESP32 microcontroller to safely control high-voltage electrical appliances. Each relay module (one for the bulb and one for the fan) is triggered by a digital signal from the ESP32's GPIO pins. When activated, the relay closes the electrical circuit for the connected appliance, turning it ON; when deactivated, it opens the circuit, turning the appliance OFF. This arrangement ensures that appliances operate safely, reliably, and precisely as per the real-time control logic dictated by human detection and temperature readings.

Chapter 4

Result

The AI Smart Room Automation system was developed with the objective of creating a real-time, intelligent automation solution capable of detecting human presence and controlling room appliances based on occupancy. This was achieved through the integration of artificial intelligence (AI), cloud-based decision-making, and IoT device control, with a focus on delivering an energy-efficient, user-friendly environment. This chapter presents the overall results and observations obtained from implementing and testing the complete system.

The system architecture comprises three main components: a camera-based sensing layer for human detection, a cloud server for data analysis and decision-making, and a microcontroller-based actuation layer for device control. The camera continuously captures live video feeds, which are processed using an AI model—primarily YOLO (You Only Look Once)—to detect the presence of humans in the room. Once human presence is identified, the information is transmitted to a cloud platform where logic and rules are applied to determine which appliances should be activated or deactivated. The decisions made in the cloud are then sent to the ESP32 microcontroller, which controls connected devices such as lights, fans, and air conditioners using communication protocols like MQTT.

The result of this multi-layered system was a functional and responsive automation solution. One of the key outcomes of the system was its ability to reduce unnecessary energy consumption by ensuring that electrical appliances remained active only when someone was present in the room. When no motion or human presence is detected for a pre-defined duration, the system automatically sends commands to turn off the appliances, contributing to significant energy savings. This automation not only demonstrates tech-

nical feasibility but also aligns with the broader goal of sustainable living.

The ESP32 microcontroller acted as the central controller in the local setup. It was configured to receive control signals from the cloud and interface directly with relays connected to appliances. The use of ESP32 proved beneficial due to its Wi-Fi capability, low power consumption, and compatibility with multiple sensor inputs and outputs. It successfully executes all received commands with negligible latency and no observed failures in communication.

Another important aspect of the system was its scalability and adaptability. The modular design allowed for easy addition of new devices or rules without disrupting the core functionality. Furthermore, the entire system was monitored and controlled through a basic web dashboard, providing users with an interface to view occupancy status. From a usability perspective, the system was intuitive and non-intrusive. Since the AI operated in the background and the control logic was entirely automated, users did not need to interact with the system directly. This seamless operation highlighted the potential for real-world adoption in smart homes, offices, or even classrooms where energy management and automation are priorities.

The AI Smart Room Automation system successfully fulfilled its intended purpose by demonstrating a working prototype capable of integrating AI-based human detection with cloud-driven decision-making and IoT device control. The results validated the effectiveness of using AI for intelligent environment control and showed how automation could lead to both convenience and conservation. While further work is possible in areas like predictive analytics, advanced learning models, and security, the current implementation provides a robust foundation for future expansion.

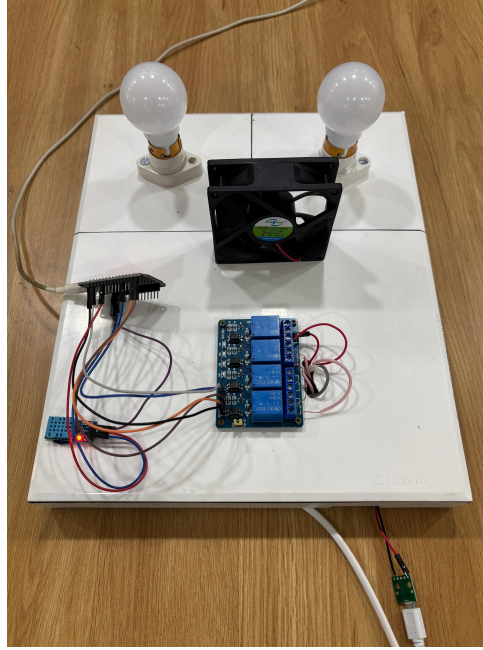


Figure 4.1: Proposed Method: Working Setup

4.1 This image shows the setup of our project. It consists of two light bulbs one represents the left side and the other the right side of the room. A fan is placed in the center of the room shown as being placed in between the two bulbs.

There are three different states in which the setup can be at a given point in time, the cases are firstly, a person is present on the left side of the room in which case the presence will be detected on the left side of the room and the lights on the left side of the room will be turned on, secondly, a person is present on the right side on the room in which case the presence will be detected on the right side of the room and the lights on the right side of the room will be turned on, and thirdly two persons are present on either side of the room in which case the presence will be detected on either sides of the room and the lights on either sides of the room will be turned on. The fan can be ON in any and/or all aforementioned cases. All the cases are shown below:

Case I: Human Present on The Left Side

In this case the human is present on the left side of the room so the lights on the left side of the room should be switched ON.

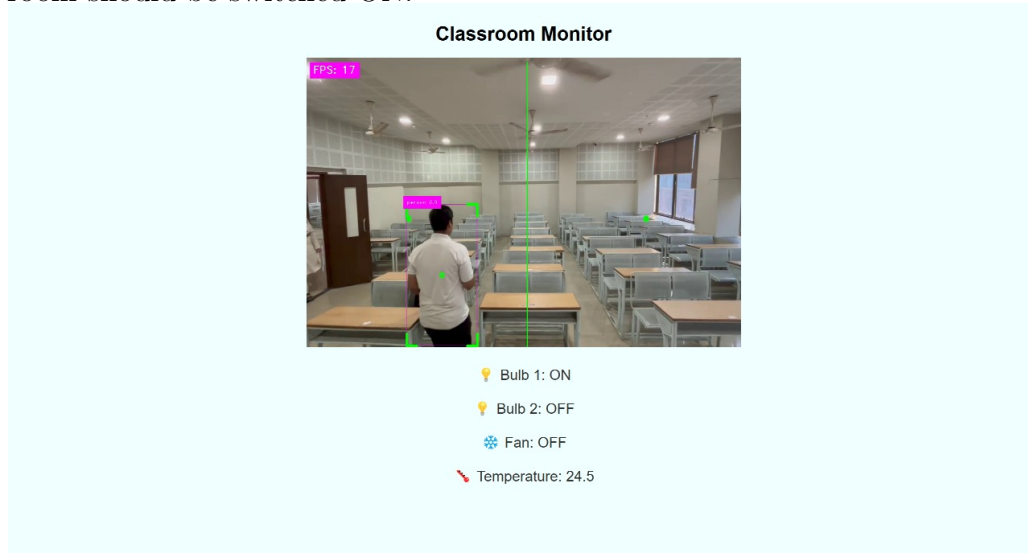


Figure 4.2: Classroom Monitor: Human Present Left Side

4.2 The human is present on the left side of the room, YOLO will detect that the human is present on the left side of the room and send the signal through cloud to ESP32, telling it to switch the left bulb ON.

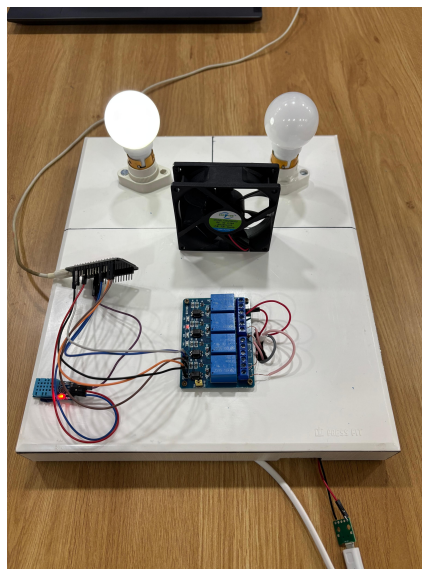


Figure 4.3: Left Devices ON

4.3 After receiving the signal send through cloud the ESP32 will then tell the relay to switch the left bulb ON. As we can see in the image the left bulb is switched ON.

Case II: Human Present on The Right Side

In this case the human is present on the right side of the room so the lights on the right side of the room should be switched ON.

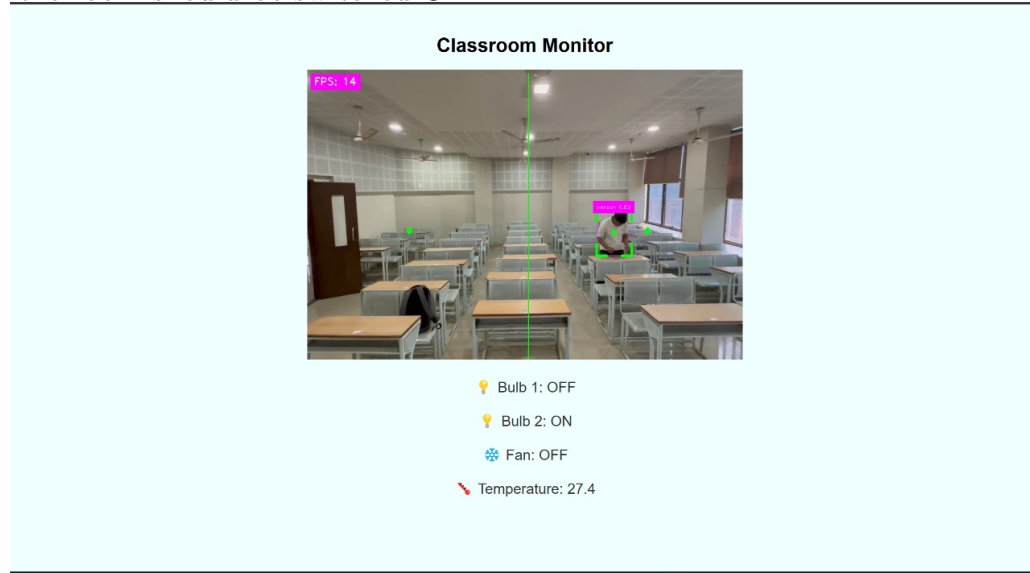


Figure 4.4: Classroom Monitor: Human Present Right Side

4.4 The human is present on the right side of the room, YOLO will detect that the human is present on the right side of the room and send the signal through cloud to ESP32, telling it to switch the right bulb ON.

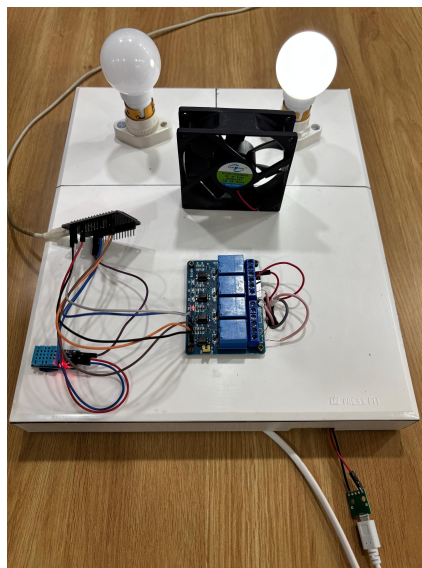


Figure 4.5: Right Devices ON

4.5 After receiving the signal send through cloud the ESP32 will then tell the relay to switch the right bulb ON. As we can see in the image the right bulb is switched ON.

Case III: Human Present on Either Sides

In this case the humans are present on the either sides of the room so the lights on either sides of the room should be switched ON and the temperature is above a set threshold meaning that the fan should also be switched on..

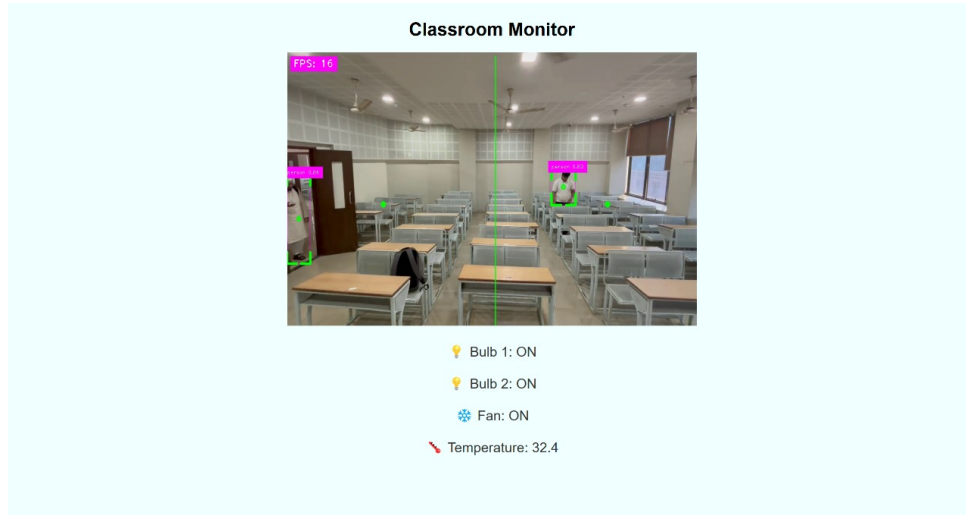


Figure 4.6: Classroom Monitor: Human Present And Temperature>28

4.6 Humans are present on either sides of the room, YOLO will detect that the human is present on either sides of the room and the DHT11 will detect that the temperature is greater than the threshold and send the signal through cloud to ESP32, telling it to switch both bulbs and the fan ON.

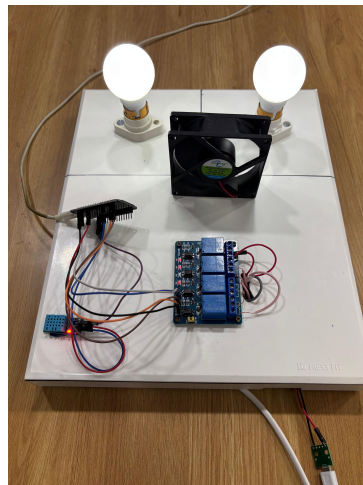


Figure 4.7: All Devices ON

4.7 After receiving the signal send through cloud the ESP32 will then tell the relay to switch the both bulbs and the fan ON. As we can see in the image both bulbs and the fan are switched ON.

Chapter 5

Conclusion

The integration of Artificial Intelligence (AI) and the Internet of Things (IoT) has brought about transformative changes in the way modern living environments are designed and managed. This project, titled *AI Smart Room Automation*, presented a comprehensive approach toward creating intelligent, responsive, and energy-efficient living spaces. The core objective of the system was to enable real-time automation of household or institutional appliances by detecting human presence and making smart decisions based on environmental inputs.

The proposed system successfully leverages computer vision technologies such as YOLO and OpenCV to detect human occupancy using a live video feed from a room. This data is transmitted to a cloud server that hosts the decision-making logic, which determines whether appliances should be turned on or off. Commands are then sent to IoT-enabled devices, ensuring seamless and intelligent control of appliances like lights, fans, and air conditioners. The architecture not only reduces the need for manual operation but also minimizes power wastage, contributing to a more sustainable and efficient energy usage model.

One of the most significant contributions of this system is its adaptability and modularity. It is designed to scale across different environments—ranging from homes and offices to classrooms, hospitals, and hotels. Moreover, it introduces a new level of convenience through remote monitoring and control via mobile or web applications. The real-time decision-making capability, combined with cloud integration and IoT connectivity, sets the stage for a smart living paradigm that is proactive rather than reactive.

In conclusion, this project is a step toward realizing the broader vision of smart cities and

intelligent environments. It illustrates how the convergence of AI and IoT can lead to practical, scalable, and human-centered automation systems. With further development, the system has the potential to redefine how people interact with their environments—making spaces not just connected, but context-aware and self-adjusting. This lays a strong foundation for future innovations in ambient intelligence and sustainable living technologies.

5.1 Application: Smart Paking Mangement

The Smart Parking Management System leverages multiple camera feeds and AI/ML algorithms to optimize parking space utilization and guide drivers to available spots with energy-efficient lighting. Upon vehicle detection at the entry barrier, the system identifies the most suitable free parking spot based on proximity, occupancy history, and configurable preferences. Once a spot is selected, guiding lights along the route illuminate sequentially, while non-essential fixtures remain dimmed at 50% intensity to conserve energy. As the vehicle traverses the lot, camera feeds and embedded sensors continuously update its position; the system dynamically maintains full illumination on the active path and reverts auxiliary lights to their low-power state.

The system architecture comprises a network of IP cameras strategically mounted throughout the parking area, streaming video to a centralized AI server. A computer vision module—employing YOLOv8 for vehicle detection and tracking—analyzes each feed to map occupied and vacant spaces in real time. A 3D occupancy map can be rendered dynamically to provide a top-down visualization of the entire facility, facilitating remote monitoring and data analytics. Upon spot assignment, the server dispatches MQTT messages to embedded lighting controllers with LED drivers, modulating individual fixtures for zone-based illumination control.

Key benefits include reduced search times for drivers, lower operational costs through adaptive lighting, and enhanced safety via clearly defined traffic pathways. Scalability is inherent: expanding to multi-level garages or large open-air lots involves simply adding cameras and lighting zones without altering the core AI/IoT framework. This solution is ideal for commercial complexes, airports, and smart cities seeking dynamic parking allocation and energy-efficient operations.

Bibliography

- [1] Bolaji Iyanu Adekunle, Ezinne C Chukwuma-Eke, Emmanuel Damilare Balogun, and Kolade Olusola Ogunsola. Machine learning for automation: Developing data-driven solutions for process optimization and accuracy improvement. *Machine Learning*, 2(1), 2021.
- [2] Ali Akram, Sagheer Abbas, Muhammad Khan, Atifa Athar, Taher Ghazal, and Husam Al Hamadi. Smart energy management system using machine learning. *Computers, Materials & Continua*, 78(1), 2024.
- [3] Arduino. Httpclient.
- [4] Arduino. Wifi. <https://docs.arduino.cc/libraries/wifi/>,.
- [5] Luigi Atzori, Antonio Iera, and Giacomo Morabito. The internet of things: A survey. *Computer networks*, 54(15):2787–2805, 2010.
- [6] Gurmu M Debele and Xiao Qian. Automatic room temperature control system using arduino uno r3 and dht11 sensor. In *2020 17th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP)*, pages 428–432. IEEE, 2020.
- [7] Python Software Foundation. Python 3 documentation. <https://docs.python.org/3/library/threading.html#module-threading>,.
- [8] Python Software Foundation. Python 3 documentation. <https://docs.python.org/3/library/json.html#module-json>,.
- [9] Eyke Hüllermeier. Prescriptive machine learning for automated decision making: Challenges and opportunities. *arXiv preprint arXiv:2112.08268*, 2021.

- [10] Ultralytics Inc. Running pre-trained yolo model in opencv. <https://docs.ultralytics.com/usage/python/>.
- [11] Peiyuan Jiang, Daji Ergu, Fangyao Liu, Ying Cai, and Bo Ma. A review of yolo algorithm developments. *Procedia computer science*, 199:1066–1073, 2022.
- [12] Sandeep Kumar and Poonam Bansal. Smart home automation using iot and ai. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, 5(2), 2019.
- [13] Alexander Maier, Andrew Sharp, and Yuriy Vagapov. Comparative analysis and practical implementation of the esp32 microcontroller module for the internet of things. In *2017 Internet Technologies and Applications (ITA)*, pages 143–148. IEEE, 2017.
- [14] Akash Malik and Hari Om. Cloud computing and internet of things integration: Architecture, applications, issues, and challenges. *Sustainable cloud and energy services: Principles and practice*, pages 1–24, 2018.
- [15] P Nagaraj, S Sathish, M Mathan Kumar, Arsalan Ahmad Khan, Ahmad Mohiuddin, and M Syed Haroon. Yolo: Human detection-based intelligent home automation using iot. In *2023 International Conference on Computer Communication and Informatics (ICCCI)*, pages 1–6. IEEE, 2023.
- [16] R Niranjana, S Arvind, M Vignesh, and S Vishaal. Effectual home automation using esp32 nodemcu. In *2022 International Conference on Automation, Computing and Renewable Systems (ICACRS)*, pages 1–5. IEEE, 2022.
- [17] Pallets. Flask. <https://flask.palletsprojects.com/en/stable/>,.
- [18] Sanguk Park. Machine learning-based cost-effective smart home data analysis and forecasting for energy saving. *Buildings*, 13(9):2397, 2023.
- [19] A Kenneth Reitz Project. Requests: Http for humans™. [https://requests.readthedocs.io/en/latest/requests-http-for-humans,](https://requests.readthedocs.io/en/latest/requests-http-for-humans,.).
- [20] Joseph Redmon and Ali Farhadi. YOLOv3: An incremental improvement. *arXiv preprint arXiv:1804.02767*, 2018.

- [21] M Senthilkumar, S Ravikrishna, M Sathiyathan, B Aiswarya, and K Ananthi. Yolo-based adaptive home automation system. In *2024 International Conference on Smart Systems for Electrical, Electronics, Communication and Computer Engineering (ICSSECC)*, pages 552–556. IEEE, 2024.
- [22] Cristina Stolojescu-Crisan, Calin Crisan, and Bogdan-Petru Butunoi. An iot-based smart home automation system. *Sensors*, 21(11):3784, 2021.
- [23] M Sujatha, N Priya, A Beno, T Blesslin Sheeba, M Manikandan, I Monica Tresa, P Subha Hency Jose, Vijayakumar Peroumal, and Sojan Palukaran Timothy. Iot and machine learning-based smart automation system for industry 4.0 using robotics and sensors. *Journal of Nanomaterials*, 2022(1):6807585, 2022.
- [24] OpenCV Team. Running pre-trained yolo model in opencv. https://docs.opencv.org/4.x/da/d9d/tutorial_dnn_yolo.html#autotoc_md463.
- [25] Juan Terven, Diana-Margarita Córdova-Esparza, and Julio-Alejandro Romero-González. A comprehensive review of yolo architectures in computer vision: From yolov1 to yolov8 and yolo-nas. *Machine learning and knowledge extraction*, 5(4):1680–1716, 2023.